SILAS R. THORNE

Principal Full Stack Architect & Engineering Lead

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PROFESSIONAL SUMMARY

Technically sophisticated and results-driven Principal Full Stack Architect with over 12 years of experience building mission-critical distributed systems, scalable SaaS platforms, and high-frequency FinTech applications. Expert in orchestrating the convergence of modern frontend frameworks (React, Next.js, TypeScript) with robust, event-driven backend architectures (Node.js, Go, Python, Rust). Distinguished for leading the digital transformation of "Nexus Global FinTech," where I transitioned a monolithic legacy system into a cloud-native microservices ecosystem, resulting in a 400% increase in transaction throughput and a $40M reduction in annual infrastructure overhead. An elite problem-solver with a proven track record of managing engineering teams of 40+ developers across global time zones. Strategic architect who bridges the gap between complex business requirements and high-performance engineering, ensuring 99.99% system availability for platforms handling over $2B in annual processing volume. Silas possesses deep expertise in cloud-native technologies (AWS, GCP), container orchestration (Kubernetes), and advanced database modeling (PostgreSQL, MongoDB, Cassandra). A champion of DevOps culture, CI/CD automation, and secure-by-design principles who excels in environments requiring high scalability and rigorous compliance.

CORE COMPETENCIES

- **Architectural Leadership:** Microservices, Event-Driven Architecture (EDA), Serverless Frameworks, Domain-Driven Design (DDD), Event Sourcing, CQRS, Hexagonal Architecture, API-First Design.
- **Frontend Excellence:** React 18+, Next.js (App Router), TypeScript, Redux Toolkit, Tailwind CSS, WebAssembly (Wasm), Performance Profiling, Web Vitals Optimization, Micro-frontends.
- **Backend Mastery:** Distributed Systems in Go (Golang) and Node.js, GraphQL Federation, gRPC, Python (FastAPI), Rust (Axum), Actor Model, Concurrency Patterns, SignalR, Socket.io.
- **Cloud & Infrastructure:** AWS (EKS, Lambda, Aurora, S3, CloudFront), Google Cloud Platform (GCP), Terraform (IaC), Multi-Cloud Strategy, Service Mesh (Istio), CloudNative Patterns.
- **Database Strategy:** SQL Optimization, NoSQL Modeling, Redis Caching Clusters, Elasticsearch, Time-Series Databases (InfluxDB), Vector DBs (Pinecone, Milvus), Data Sharding.
- **DevOps & CI/CD:** Kubernetes Administration, GitHub Actions, Docker, Prometheus/Grafana Observability, Chaos Engineering, ArgoCD, Helm, Infrastructure Monitoring.
- **Engineering Leadership:** Agile/Scrum Mastery, Technical Mentorship, OKR Strategic Planning, M&A Technical Due Diligence, Engineering Culture Transformation.
- **Security & Compliance:** OWASP Top 10 Mitigation, OAuth2/OIDC, JWT Security, PCI-DSS Implementation, Zero-Trust Architecture, SOC2 Readiness, Data Sovereignty.

TECHNICAL SKILLS

Frontend Engineering Ecosystem

- **Frameworks & Libraries:** React (Context API, Hooks, Server Components, Suspense), Next.js (App Router, ISR, SSR, SSG), Vue.js 3 (Composition API, Pinia), Angular 14+, SvelteKit, Preact.
- **State Management:** Redux Toolkit (Saga, Thunk), Zustand, Recoil, TanStack Query (React Query), Jotai, XState (Finite State Machines), MobX.
- **Styling & UI Components:** Tailwind CSS, CSS Modules, Styled Components, Framer Motion, Material UI (MUI), Shadcn/UI, Radix UI, Headless UI, DaisyUI.
- **Testing & Tooling:** Jest, Vitest, Cypress (E2E), Playwright, React Testing Library, Storybook, NX Monorepo, Turborepo, Webpack, Vite, Esbuild, Rollup.

Backend & API Development

- **Languages:** Go (Golang - Gin, Echo), Node.js (TypeScript, NestJS, ESM), Rust (Tokio, Axum, Serde), Python 3.11+ (FastAPI, Pydantic, Django), Java (Spring Boot 3), C++ (Low-level hooks).
- **API Standards:** GraphQL (Apollo Federation, Subgraphs), RESTful APIs (HATEOAS), gRPC (Protocol Buffers), WebSockets (Socket.io), tRPC, OpenAPI/Swagger, JSON:API.
- **Messaging & Event Streaming:** Apache Kafka (Kafka Streams, Connect), RabbitMQ, Amazon SQS/SNS, Redis Pub/Sub, NATS, Google Pub/Sub, EventBridge.

Infrastructure & Cloud Platforms

- **AWS Cloud Stack:** EC2, EKS (Fargate), Lambda, S3, RDS (Aurora/PostgreSQL), DynamoDB, CloudFront, Route53, IAM, VPC, Secrets Manager, Kinesis, AppSync.
- **Containerization:** Docker, Podman, Kubernetes (Helm, Kustomize), ArgoCD (GitOps), Istio Service Mesh, Linkerd, Cilium, Containerd.
- **Infrastructure as Code (IaC):** Terraform (HCL), AWS CDK, Pulumi, CloudFormation, Ansible, Packer, Crossplane.

Data Management & Analytics

- **Relational Databases:** PostgreSQL (PostGIS, Citus), MySQL, Microsoft SQL Server, CockroachDB (Distributed SQL), MariaDB.
- **NoSQL & Specialty:** MongoDB, Cassandra (CQL), DynamoDB, Couchbase, Neo4j (Cypher), Pinecone (Vector), Weaviate, ClickHouse.
- **Caching & Search:** Redis (Cluster, Sentinel, Redlock), Memcached, Elasticsearch (ELK Stack), Meilisearch, Algolia, Solr.

PROFESSIONAL EXPERIENCE

Principal Full Stack Architect | Nexus Global FinTech | 2020 – Present

Boston, MA (Premier Digital Banking & Payments Infrastructure | Revenue: $4.2B)

Strategic Leadership & Architectural Oversight: Appointed as the lead architect to modernize the core payment processing engine and consumer-facing web applications. Oversee a technical organization of 45 engineers divided into 5 autonomous squads. Responsible for the technology roadmap, architectural governance, and vendor management for a multi-cloud infrastructure handling mission-critical financial data.

- **Legacy Decoupling Initiative:** Led the 24-month transition from a PHP/jQuery monolith to a React/Next.js frontend and Go-based microservices. Achieved a 60% reduction in page load times and a 35% improvement in developer velocity through a unified monorepo strategy (TurboRepo) and automated code generation tools.
- **High-Scale API Gateway:** Architected a custom GraphQL Federation gateway using Apollo Router that aggregates data from 40+ subgraphs. Successfully handled a peak of 120,000 requests per second during the 2023 Black Friday period with zero downtime and sub-50ms latency.
- **Infrastructure Cost Optimization:** Implemented a "Serverless First" strategy for non-core services using AWS Lambda and Aurora Serverless. Reduced monthly cloud spend by $350,000 while improving regional scalability in EMEA and APAC via edge computing deployments.
- **Internal Design System:** Spearheaded the creation of "Nexus-UI," a comprehensive React component library and Figma tokens system. Consistently reduced frontend development time for new features by 45% across the entire organization.
- **Observability & Resilience:** Deployed a full-stack observability suite using OpenTelemetry, Honeycomb, and Datadog. Reduced Mean Time to Recovery (MTTR) by 70% by implementing automated circuit breakers and localized fallback strategies in the frontend.
- **Security Transformation:** Directed the migration to a modern AuthN/AuthZ stack using Okta and custom OPA (Open Policy Agent) rules. Ensured 100% compliance with updated PCI-DSS 4.0 standards and implemented mTLS for inter-service communication.
- **M&A Due Diligence:** Conducted technical audits for 3 strategic acquisitions, identifying over $5M in potential technical debt and facilitating the seamless integration of their engineering teams into the Nexus ecosystem.

Staff Software Engineer (Lead) | QuantEdge Systems | 2016 – 2020

New York, NY (High-Frequency Trading & Quantitative Analytics)

Technical Leadership: Served as the lead developer for the "AlphaStream" trading dashboard, a real-time analytics platform used by institutional traders to monitor and execute high-volume market orders in highly volatile environments.

- **Real-Time Data Visualization:** Engineered a high-performance WebGL-based charting engine using Three.js and D3.js, capable of rendering 1 million data points at 60fps. Reduced client-side memory usage by 40% through custom binary data serialization and Web Workers.
- **Low-Latency Messaging Hub:** Built a WebSocket-based data bridge in Node.js that streamed market data with sub-10ms latency to 10,000+ concurrent clients. Optimized the garbage collection cycles and memory heap management to prevent "stuttering" during high volatility.
- **Distributed Computation Engine:** Developed a Python-based backend service for complex risk-modeling using Celery and Redis. Scaled the system to handle 50,000 concurrent simulations, providing traders with real-time "Value at Risk" (VaR) calculations.
- **Frontend State Management:** Migrated a massive legacy Redux codebase to Redux Toolkit and RTK Query, eliminating 30,000 lines of boilerplate code and improving type safety across the frontend using advanced TypeScript utility types.
- **Automated Quality Assurance:** Established a rigorous E2E testing pipeline using Cypress and

a custom "visual regression" tool that prevented 15+ critical UI bugs from reaching production in the first year.

- **Kafka Migration:** Led the migration from a legacy RabbitMQ setup to a distributed Kafka cluster, improving data persistence and allowing for high-throughput replayability of market events for regulatory audits.
- **Mentorship:** Formally mentored 8 senior engineers, 4 of whom were promoted to Staff level within 18 months of coaching through structured career pathing and technical drills.

Senior Full Stack Developer | CloudArc Innovations | 2013 – 2016

San Francisco, CA (Cloud-Based Workflow Automation SaaS)

Role Overview: Part of the early engineering team responsible for building "FlowState," a collaborative project management tool. Focused on real-time collaborative features and multi-tenant database architecture.

- **Collaborative Engine:** Implemented Operational Transformation (OT) logic to enable real-time, multi-user document editing similar to Google Docs. Reduced synchronization conflicts by 95% in high-concurrency scenarios through conflict-free replicated data types (CRDTs).
- **Multi-Tenant Database Design:** Architected a "Schema-per-Tenant" PostgreSQL strategy that ensured strict data isolation for enterprise clients while maintaining high query performance through shared indexing strategies and pg_partman.
- **API Ecosystem:** Designed and documented the public REST API (Swagger/OpenAPI), which grew to support 500+ third-party integrations (Zapier, Slack, Salesforce). Managed the versioning strategy to ensure backward compatibility.
- **Performance Tuning:** Optimized Ruby on Rails backend by implementing advanced caching layers and database connection pooling, resulting in a 50% reduction in average API response times and improved throughput.
- **SOC2 Compliance:** Co-led the technical efforts to achieve SOC2 Type II certification, implementing encrypted storage at rest/transit and robust audit logging for all administrative actions.

Full Stack Developer | SparkWorks Creative | 2011 – 2013

Austin, TX (Boutique Software Agency)

- **Client Success:** Delivered 12+ custom web applications for clients ranging from startups to Fortune 500 companies (e.g., Fictional Client "NorthStar Logistics"), maintaining a 100% on-time delivery record.
- **Mobile First Strategy:** Led the transition to responsive design using Bootstrap 2/3 and media queries, ensuring seamless user experiences across mobile devices during the early smartphone boom.
- **CMS Customization:** Developed high-performance custom themes and plugins for WordPress and Drupal, handling high-traffic marketing campaigns for entertainment brands with millions of hits per day.
- **Email Marketing Engine:** Built a custom Ruby-based email delivery platform that handled 1M+ monthly sends with 99% deliverability rate through IP warming and reputation management.

KEY PROJECTS PORTFOLIO

Project Singularity: The Next-Gen Payments Core

Role: Lead Architect | **Stack:** Go, PostgreSQL, Kafka, React, AWS.

- **Objective:** Replace a 15-year-old COBOL-based ledger system with a modern, immutable distributed ledger capable of handling global settlement.
- **Strategy:** Utilized an Event-Sourcing pattern to ensure perfect auditability and data integrity. Built the frontend with React Server Components for near-instant load times for administrative staff.
- **Outcome:** Enabled sub-second payment settlement times. Reduced database storage requirements by 30% through efficient event pruning and snapshotting. Successfully handles $500M in daily transaction volume.

The Aether Protocol: Secure Decentralized Identity

Role: Technical Lead | **Stack:** TypeScript, Rust (Wasm), Node.js, DynamoDB.

- **Objective:** Build a ”Zero-Knowledge” identity verification system for sensitive financial transactions to comply with international KYC/AML laws.
- **Strategy:** Leveraged WebAssembly to run cryptographic proofs directly in the browser, ensuring user data never leaves the device in an unencrypted state.
- **Outcome:** Successfully integrated with 5 major European banks. Cited as a ”Security Innovation of the Year” in an internal FinTech journal for its privacy-preserving architecture.

Project Hydra: Real-Time Fraud Detection Engine

Role: Principal Architect | **Stack:** Python, FastAPI, Redis, Apache Flink, React.

- **Objective:** Develop an ML-powered fraud detection system that analyzes transactions in real-time to identify anomalous patterns.
- **Strategy:** Implemented a stream processing pipeline using Flink to analyze transaction patterns against pre-trained models. Used Redis for millisecond-latency feature store access.
- **Outcome:** Reduced fraudulent transaction approval rate by 42%, saving the company $12M in the first fiscal year of deployment without increasing false positives.

Lumina Web Analytics: High-Performance Tracking

Role: Backend Architect | **Stack:** Go, ClickHouse, Kafka, Next.js.

- **Objective:** Build a real-time web analytics platform capable of processing 1 billion events per month for enterprise SaaS clients.
- **Implementation:** Developed a high-throughput ingestion worker in Go that writes to Kafka. Used ClickHouse as the primary OLAP database for sub-second analytical queries.
- **Outcome:** Replaced a costly third-party solution, saving the client $200,000 annually in SaaS fees while providing 10x faster data visibility.

LEADERSHIP & TEAM MANAGEMENT

- **Engineering Culture:** Established a "Documentation-First" culture, mandating RFCs (Request for Comments) for all significant architectural changes. Resulted in a 20% reduction in technical debt accrual and significantly improved cross-team communication.
- **Hiring & Growth:** Personally interviewed and hired 50+ engineers. Developed a "Career Progression Matrix" that provided clear growth paths for Individual Contributors and Managers, leading to a 92% engineering retention rate over 4 years.
- **Agile Transformation:** Migrated the organization from "Waterfall-Agile" to a true Continuous Delivery model, moving from bi-weekly releases to 10+ production deployments per day through automated canary testing.
- **Mentorship:** Conducted weekly "Level-Up" sessions for senior and staff engineers, focusing on system design, distributed systems theory, and leadership soft skills.
- **Diversity & Inclusion:** Spearheaded a diverse hiring initiative that increased underrepresented minority representation in engineering by 25% in 2 years through blind resume screening and targeted outreach.

PROCESS IMPROVEMENTS & AUTOMATION

- **Automated Dependency Management:** Implemented Renovate Bot with automated "Green-Build" merging, keeping 98% of project dependencies up to date with zero manual intervention across 200+ repositories.
- **K8s Cost Management:** Developed a custom tool that monitors idle Kubernetes pods and scales down dev/staging environments outside of working hours, saving $15,000 per month.
- **Self-Service Infrastructure:** Built a "Developer Portal" using Backstage.io that allows engineers to provision new microservices (including CI/CD, K8s namespaces, and Databases) in under 5 minutes.
- **Visual Regression Testing:** Automated UI testing for 500+ unique web views using Playwright's visual comparison feature, reducing QA cycle time by 3 days per release.
- **Automated Compliance Evidence:** Built a tool to automatically export security logs and configuration states into Drata, ensuring continuous SOC2 readiness and zero audit findings.
- **Canary Release Automation:** Integrated Flagger with Istio to automate canary rollouts, automatically rolling back deployments if error rates exceed a 0.5% threshold.

EDUCATION & CERTIFICATIONS

Master of Science in Computer Science (Distributed Systems) | *Massachusetts Institute of Technology (MIT)*, Cambridge, MA
Research Focus: High-Availability Consistency Protocols (Paxos, Raft). GPA: 4.0/4.0
Bachelor of Science in Software Engineering | *University of Texas at Austin*, Austin, TX
Honors: Graduated Magna Cum Laude; President of the Association for Computing Machinery (ACM) Student Chapter.
Professional Certifications:

- AWS Certified Solutions Architect – Professional
- CKA: Certified Kubernetes Administrator (Cloud Native Computing Foundation)

- Google Cloud Professional Cloud Architect
- Terraform Associate Certification (HashiCorp)
- Offensive Security Certified Professional (OSCP) – Specialized in secure coding and penetration testing.
- Professional Scrum Master II (PSM II)

TOOLS & TECHNOLOGIES REFERENCE

Infrastructure: Docker, Kubernetes, Terraform, Helm, Istio, Linkerd, Consul, Cloudflare (Edge/WAF), Vault, Nginx, Traefik, Kong API Gateway.
Observability: Prometheus, Grafana, Jaeger (Tracing), ELK Stack (Elasticsearch, Logstash, Kibana), Sentry, Honeycomb, New Relic, BetterStack.
Workflow: Jira, Confluence, Linear, Slack, Notion, Miro (System Diagramming), Lucidchart, Figma (for design-to-code handoff).
Development: VS Code, Neovim, JetBrains Suite (IntelliJ, WebStorm, GoLand), Git (GitHub/GitLab), Postman, Insomnia, Charles Proxy, Fiddler.
Compliance & Security: Vanta, Drata, Snyk, SonarQube, Prisma Cloud, Aqua Security, HashiCorp Vault, OneTrust, Burp Suite.

PUBLICATIONS & CONFERENCES

Keynote: "Scaling Next.js to 100M Users: Architectural Hard Lessons," *React Conf 2023*.
Speaker: "Go Microservices in a Polyglot World: Navigating the Complexities," *GopherCon 2022*.
Publication: "The Monolith to Microservices Journey: A Financial Perspective," *Modern Software Architect Magazine*, 2022.
Publication: "Optimizing React Performance for High-Frequency Data Streams," *Frontend Masters Blog*, 2021.
Workshop Lead: "Infrastructure as Code with Terraform and AWS for Scale," *AWS re:Invent 2021*.
Panelist: "The Future of Full Stack: Is Rust the Successor to Node.js?," *Web Summit 2020*.
Publication: "Distributed Tracing in a Serverless World: Lessons from Production," *ACM Queue*, 2019.

ARCHITECTURAL PHILOSOPHY

The Convergence of Edge and Cloud: I believe the next frontier of web development lies in moving intelligence closer to the user. My current research focuses on leveraging Edge Computing to minimize latency for global financial transactions while maintaining strong eventual consistency across disparate regions.

Pragmatic Engineering: Technology should never be chosen based on industry trends. I champion a philosophy where we use the right tool for the job—whether that means a robust Rust-based microservice for safety-critical logic or a simple Serverless function for rapid prototyping.

Human-Centric Leadership: The most scalable thing in an organization isn't the software—it's the people. My ultimate goal is to build engineering teams that are empowered, curious, and relentless in their pursuit of excellence through shared ownership and a blame-free post-mortem culture.